

Number theoretic transform (NTT) accelerator

Basic Information		
IP Name		Number theoretic transform (NTT) accelerator
Licensing		Open-source (GPLv3.0)
Target FPGA or ASIC		
Design Language		
Architecture		
Type		
Num Clock Domains		1
Control Interface	ISAExtension	Y
Control Interface	Memory Mapped	N
Control Interface	Protocol	CV-X-IF
Memory Interface	Interface Protocol	CV-X-IF
Memory Interface	Cached	N
Memory Interface	IOMMU	N
Interrupts		N
Microarchitecture		
Parametrization	Parametric Num Units	N
Parametrization	Parametric Config	N
Programmability	Programmable Cores	N
Programmability	ISAExtension	RISC-V(RV32/64IMC-zicsr)
Software		
Compiler	Requires Specialized Compiler	Y
Hardware Abstraction Layer	HAL Functions	N/A
highLevelAPI	Has High Level API	N
highLevelAPI	SDK Repository	N/A
highLevelAPI	Has Domain Specific Compiler	N
Integration		
IP Distribution	Manifest Type	Bender.yml
IP Distribution	Standalone Simulation	Y
IP Distribution	Standalone Simulation SW Require- ments	QuestaSim
IP Distribution	Integration Docs Or Examples	https://github.com/alex2kameboss/MatrixAcceleratorDemo.git
Synthesis	Is Synthesizable	Y
Synthesis	FPGA Synthesis Examples	Y (Bender and custom scripts)
Synthesis	ASIC Synthesis Examples	N
Simulation	Closed Source Simulation	Y (QuestaSim)

Simulation	Open Source Simulation	N
Evaluation	PPA Results Available	Y, synthesis results, runtime